

φ 2 AQM (S18)

I
Scattering

Current Conservation (non-rel. QM) $H = \frac{p^2}{2m} + V$

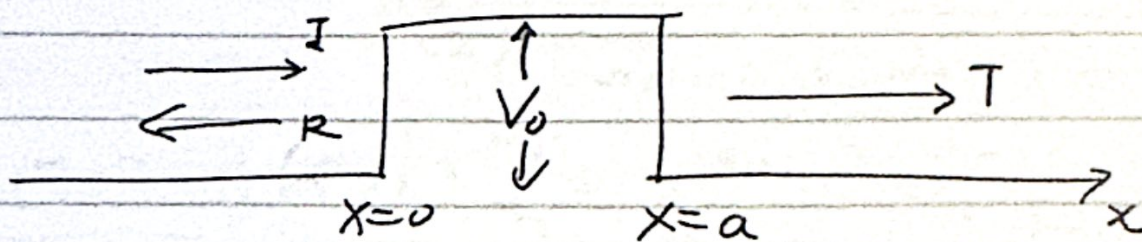
$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \underline{j} \quad (i\hbar \frac{\partial \psi}{\partial t} = H\psi)$$

where $\rho = \psi^* \psi$ $\underline{j} = -\frac{i\hbar}{2m} (\psi^* \nabla \psi - \nabla(\psi^*) \psi)$

for plane wave $\psi = A e^{-i(Et - p \cdot x)/\hbar}$

$$\underline{j} = |A|^2 \frac{p}{m}$$

Scattering in 1D



$$\psi_{x < 0} = e^{-i(\omega t - kx)} + r e^{-i(\omega t + kx)}$$

$$\psi_{0 < x < a} = A e^{-i(\omega t - k'x)} + B e^{-i(\omega t + k'x)}$$

$$\psi_{x > a} = t e^{-i(\omega t - kx)}$$

$$\frac{\hbar^2 k^2}{2m} + V_0 = E = \frac{\hbar^2 k^2}{2m}$$

to satisfy Schrödinger's eqn.

b.c.:

match ψ & $\frac{\partial \psi}{\partial x}$ at $x=0, a$.

to solve for A, B, ρ, T .

3D

Def $\delta n = f \frac{d\sigma}{d\Omega} \delta\Omega$

differential cross-section.

$$\frac{d\sigma}{d\Omega} = \frac{\delta n}{f \delta\Omega}$$

$$\left| \frac{d\sigma}{d\Omega} = |f_k(\theta, \phi)|^2 \right|$$

assuming the expansion

$$\psi = e^{ikz} + \frac{e^{ikr}}{r} f_k(\theta, \phi) (1 + O(r^{-1}))$$

$$\underline{j} = \underline{\hat{z}} \frac{\hbar k}{\mu} + \frac{\underline{\hat{r}}}{r^2} \frac{\hbar k}{\mu} |f_k(\theta, \phi)|^2 + O(r^{-3})$$

+ ...
interference

Dirac Delta

Convention $\int_{-\infty}^{\infty} \delta(x) dx = 1$

$$\int_{-\infty}^{\infty} \delta(x) g(x) dx = g(0)$$

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk$$

Green's functions

$$(\nabla^2 + k^2) G(\underline{x}) = \delta^3(\underline{x})$$

can write

$$\psi(\underline{x}) = \psi_0(\underline{x}) + \int d^3\underline{x}' G(\underline{x} - \underline{x}') U(\underline{x}') \psi(\underline{x}')$$

such that $(\nabla^2 + k^2) \psi_0(\underline{x}) = 0$

$$(\nabla^2 + k^2) \psi(\underline{x}) = U(\underline{x}) \psi(\underline{x})$$

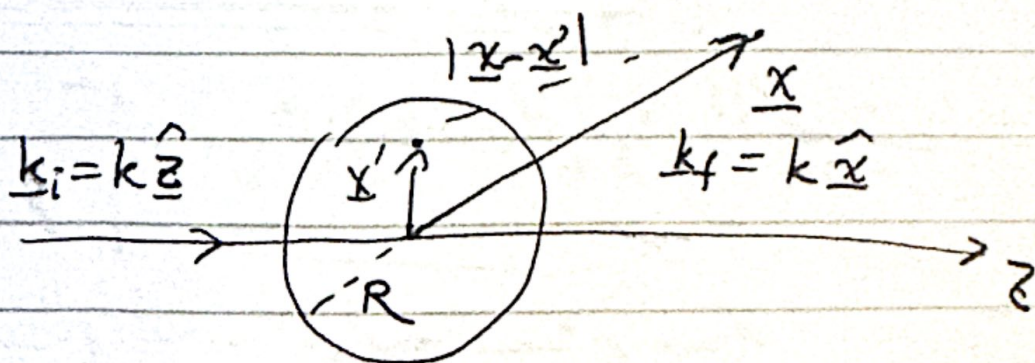
choose $\psi_0(\underline{x}) = e^{ikz}$

$$G(\underline{x}) = -\frac{e^{ik|\underline{x}|}}{4\pi|\underline{x}|} \quad (\text{shown later})$$

Born Approx.

1st order:

$$\psi^{(1)}(\underline{x}) = \psi_0(\underline{x}) + \int d^3\underline{x}' G(\underline{x} - \underline{x}') U(\underline{x}') \psi_0(\underline{x}')$$



$$\psi_{\text{scatt}}^{(1)}(\underline{x}) = - \int d^3x' \frac{e^{ik|\underline{x}-\underline{x}'|}}{4\pi|\underline{x}-\underline{x}'|} V(\underline{x}') e^{i\vec{k}\cdot\underline{x}'} \quad \underline{\hat{z}} \cdot \underline{x}'$$

$$\left(\text{where } |\underline{x}-\underline{x}'| = \sqrt{(\underline{x}-\underline{x}') \cdot (\underline{x}-\underline{x}')} \right. \\ \left. \approx |\underline{x}| - \hat{\underline{x}} \cdot \underline{x}' \right)$$

$$= - \frac{e^{ik|\underline{x}|}}{4\pi|\underline{x}|} \int d^3x' e^{i(\underline{k}_i - \underline{k}_f) \cdot \underline{x}'} V(\underline{x}')$$

$$\text{where } \left(-\frac{\hbar^2 \nabla^2}{2\mu} + V \right) \psi = E \psi$$

⇓

$$(\nabla^2 + k^2) \psi = U \psi$$

$$(U = \frac{2\mu}{\hbar^2} V)$$

$$\text{Thus } f_k(\theta, \phi) = - \frac{1}{4\pi} \int d^3x' e^{i(\underline{k}_i - \underline{k}_f) \cdot \underline{x}'} V(\underline{x}') \\ = - \frac{\mu}{2\pi\hbar^2} \int d^3x' e^{i(\underline{k}_i - \underline{k}_f) \cdot \underline{x}'} V(\underline{x}')$$

$$\frac{d\sigma}{d\Omega} = \left| \frac{\mu}{2\pi\hbar^2} \right|^2 \left| \int d^3x' e^{i(\underline{k}_i - \underline{k}_f) \cdot \underline{x}'} V(\underline{x}') \right|^2$$

Fourier transform of scatt. potential.

Lippmann-Schwinger Eqn

Use $(E - H_0)$
operator!

$$|\psi_{\pm}\rangle = |\phi\rangle + (E - H_0 \pm i\epsilon)^{-1} V |\psi_{\pm}\rangle$$

$$\text{where } \begin{cases} H_0 |\phi\rangle = E |\phi\rangle \\ H_0 |\psi_{\pm}\rangle = (E - V) |\psi_{\pm}\rangle + O(\epsilon) \end{cases}$$

take $\langle x | \psi_{\pm} \rangle$

$$= \langle x | \phi \rangle + \langle x | (E - H_0 \pm i\epsilon)^{-1} \int d^3 x' |x'\rangle \langle x' | V | \psi_{\pm} \rangle$$

$$= \langle x | \phi \rangle + \int d^3 x' \langle x | (E - H_0 \pm i\epsilon)^{-1} | x' \rangle V(x') \langle x' | \psi_{\pm} \rangle$$

$$\text{so } G_{\pm}(x-x') = \frac{\hbar^2}{2m} \langle x | (E - H_0 \pm i\epsilon)^{-1} | x' \rangle$$

G_+ outgoing spherical wave

G_- ingoing ...

T operator

$$V |\psi_{+}\rangle = T |\phi\rangle$$

$$T |\phi\rangle = V |\phi\rangle + V (E - H_0 + i\epsilon)^{-1} T |\phi\rangle$$

$$\underline{T = V + V (E - H_0 + i\epsilon)^{-1} T}$$

Green's function calculation

$$G_{\pm}(\underline{x} - \underline{x}') = \frac{\hbar^2}{2\mu} \int d^3p' \langle \underline{x} | (\mathcal{E} - H_0 + i\epsilon)^{-1} | \underline{p}' \rangle \langle \underline{p}' | \underline{x}' \rangle$$

$$= \frac{\hbar^2}{2\mu} \frac{1}{(2\pi\hbar)^3} \int d^3p' e^{i \frac{\underline{p}' \cdot (\underline{x} - \underline{x}')}{\hbar}} (\mathcal{E} - \frac{p'^2}{2\mu} \pm i\epsilon)^{-1}$$

$$\underline{p}' = \hbar \underline{k}' \quad \& \quad \mathcal{E} = \frac{\hbar^2 k^2}{2\mu}$$

$$G_{\pm}(\underline{x} - \underline{x}') = \int \frac{d^3k'}{(2\pi)^3} \frac{e^{i \underline{k}' \cdot (\underline{x} - \underline{x}')}}{k^2 - k'^2 \pm i\epsilon}$$

$$= \int_0^{\infty} \frac{k'^2 dk'}{(2\pi)^3} \int_0^{2\pi} d\phi' \int_{-1}^1 d\cos\theta' \frac{e^{i k' |\underline{x} - \underline{x}'| \cos\theta'}}{k^2 - k'^2 \pm i\epsilon}$$

$$= -\frac{i}{(2\pi)^2} \frac{1}{2|\underline{x} - \underline{x}'|} \int_{-\infty}^{\infty} \frac{k' dk'}{k^2 - k'^2 \pm i\epsilon} \left(e^{-i k' |\underline{x} - \underline{x}'|} - e^{i k' |\underline{x} - \underline{x}'|} \right)$$

(by complex analysis)

$$G_{+}(\underline{x} - \underline{x}') = -\frac{i}{(2\pi)^2} \frac{1}{2|\underline{x} - \underline{x}'|} (2\pi i) \left(\frac{1}{2} e^{-i(-k)|\underline{x} - \underline{x}'|} \overset{\text{C.W.}}{\uparrow} (-1) - \frac{1}{2} e^{i k |\underline{x} - \underline{x}'|} \right)$$

$$= -\frac{1}{4\pi|\underline{x} - \underline{x}'|} e^{i k |\underline{x} - \underline{x}'|}$$

Green's
function

$$\underline{G(x-x')} = -\frac{1}{4\pi|x-x'|} e^{-ik|x-x'|}$$

S-matrix

$$S_{fi} = \langle p_f | S | p_i \rangle = \langle p_f | (1 + (E_{p_i} - E_{p_f} + i\varepsilon)^{-1} T) | p_i \rangle$$

Optical Theorem

$$\begin{aligned} \sigma_{\text{tot}} &= \int d\Omega |f(k_f, k_i)|^2 \\ &= \frac{4\pi}{k_i} \text{Im} \left\{ f(k_i, k_i) \right\} \end{aligned}$$

Bound state

$E > 0 \rightarrow$ Continuum of scatt states $|\phi\rangle$

$E < 0 \rightarrow$ discrete bound states $|n\rangle$

(eigenval.)

if $S(k)$ has a positive imaginary pole

at $k = ik^*$ ($k^* > 0$)

bound state: $E = -\frac{\hbar^2 k^{*2}}{2\mu}$

II

Relativistic QM

KG & Dirac

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial t^2} = \underbrace{\left(-\frac{\hbar^2}{2m} \nabla^2 + m^2 \right)}_{p^2 + m^2} \psi \quad \text{K-G}$$

plane-wave soln $\psi(x, t) = e^{-i(Et - p \cdot x)/\hbar} \psi_0$

general soln $\psi(x, t)$
 $= a_p e^{-i(E_p t - p \cdot x)/\hbar} + b_p e^{-i(E_p t - p \cdot x)/\hbar}$

where $E_p = +\sqrt{p^2 + m^2}$

(can have arbitrarily negative energy)

Conservation law

$$\rho = \frac{i\hbar}{2m} \left(\psi^* \frac{\partial \psi}{\partial t} - \frac{\partial \psi^*}{\partial t} \psi \right)$$

$$\underline{j} = -\frac{i\hbar}{2m} \left(\psi^* \nabla \psi - (\nabla \psi^*) \psi \right)$$

- ρ is not positive-definite, more of 'charge density', not prob.

Conserved quantities

$$\beta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

$$\partial_\mu = \left(\frac{\partial}{\partial t}, \nabla \right) \rightarrow \partial^\mu = \left(\frac{\partial}{\partial t}, -\nabla \right)$$

K-G eqn $(\partial^\mu \partial_\mu + m^2) \psi = 0$

4-current $j^\mu = (\rho, \underline{j})$

$$= \frac{i}{2m} (\psi^* \partial^\mu \psi - (\partial^\mu \psi^*) \psi)$$

K-G & EM field $(A_\mu = (V, -\underline{A}))$

Minimal coupling $\partial_\mu \rightarrow \underbrace{\partial_\mu + ieA_\mu}_{D_\mu}$

KG eqn:

$$\boxed{(D^\mu D_\mu + m^2) \psi = 0}$$

Gauge-transformation

$$\begin{cases} \psi(x) \rightarrow e^{ie\alpha(x)} \psi(x) \\ A_\mu(x) \rightarrow A_\mu(x) - \partial_\mu \alpha(x) \end{cases}$$

leaves KG eqn invariant.

① Constant & non-zero scalar potential V

$$-\left(\frac{\partial}{\partial t} + ieV\right)^2 \psi = (-\nabla^2 + m^2) \psi$$

Ansatz: plane-wave $\psi(x, t) = e^{-i(Et - \underline{p} \cdot \underline{x})} \psi_0$

$$(E - eV)^2 = \underline{p}^2 + m^2$$

$$\rightarrow E = eV \pm E_p \quad (\text{particle \& anti-particle})$$

① spherically symmetric potential

$$\left(i\frac{\partial}{\partial t} - eV(r)\right)^2 \psi = (-\nabla^2 + m^2) \psi$$

where $-\nabla^2 = -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \underline{L}^2$

→ stationary eigenstates:

$$\psi = e^{-iEt} Y_{l,m}(\theta, \phi) R(r)$$

with $\underline{L}^2 Y_{l,m} = l(l+1) Y_{l,m}$

$$L_z Y_{l,m} = m Y_{l,m}$$

$$-l \leq m \leq l, \quad l = 0, 1, 2, 3, \dots$$

K-G: 2nd-order in time

Dirac: 1st-order in time only

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi = (c \underline{\alpha} \cdot \hat{\underline{p}} + mc^2 \beta) \psi$$

Apply $\psi = e^{-\frac{i}{\hbar}(Et - \mathbf{p} \cdot \mathbf{x})} \psi_0$

$$E\psi_0 = \hat{H}\psi_0 = (c\boldsymbol{\alpha} \cdot \hat{\mathbf{p}} + mc^2\beta)\psi_0$$

$$\rightarrow E^2 = c^2(\boldsymbol{\alpha} \cdot \mathbf{p})^2 + mc^3(\boldsymbol{\alpha} \cdot \mathbf{p}\beta + \beta\boldsymbol{\alpha} \cdot \mathbf{p}) + m^2c^4\beta^2$$

$$\Downarrow$$

$$\begin{cases} \alpha_i \alpha_j + \alpha_j \alpha_i = 2\delta_{ij} \mathbb{1} \\ \alpha_i \beta + \beta \alpha_i = 0 \\ \beta^2 = \mathbb{1} \end{cases} \quad (\mathbb{1} \text{ is } 4 \times 4)$$

One choice:

$$\beta = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} \quad \alpha_i = \begin{pmatrix} -\sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix}$$

$$(\mathbf{p} \cdot \boldsymbol{\alpha} + m\beta)\psi_0 = E\psi_0$$

$\downarrow$

Dirac Spinor

(4 components, not a 4-vector)

From

$$i\frac{\partial \psi}{\partial t} = (-i\nabla \cdot \boldsymbol{\alpha} + m\beta)\psi$$

one can check

$$\frac{\partial(\psi^\dagger \psi)}{\partial t} + \nabla \cdot \underline{j} = 0 \quad \checkmark$$

still charged density

$$\hookrightarrow \rho = \psi^\dagger \psi \quad \& \quad \underline{j} = \psi^\dagger \boldsymbol{\alpha} \psi$$

Angular Momentum & Spin

$$\underline{S} = \frac{\hbar}{2} \begin{pmatrix} \sigma & 0 \\ 0 & \sigma \end{pmatrix}$$

$$\left[\underbrace{L_i + S_i}_{J_i}, H \right] = 0$$

$\psi^S \rightarrow \text{spin}$
 $E \rightarrow \text{energy}$

$$H \psi_+^{\pm} = +m \psi_+^{\pm}$$

$$S_z \psi_+^{\pm} = \pm \frac{\hbar}{2} \psi_+^{\pm}$$

$$H \psi_-^{\pm} = -m \psi_-^{\pm}$$

$$S_z \psi_-^{\pm} = \pm \frac{\hbar}{2} \psi_-^{\pm}$$

General form of Dirac Eqn

$$0 = (-i\beta \frac{\partial}{\partial t} - i\beta \underline{\alpha} \cdot \nabla + m) \psi$$

$$\boxed{0 = (-i\gamma^\mu \partial_\mu + m) \psi}$$

$$\gamma^\mu = (\beta, \beta \underline{\alpha})$$

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$$

Momentum eigenvalues:

$$\psi = e^{-ip_\mu x^\mu} \psi_0(p)$$

$$(-p_\mu \gamma^\mu + m) \psi_0(p) = 0$$

$$\rightarrow (-\not{p} + m) \psi_0(p) = 0$$

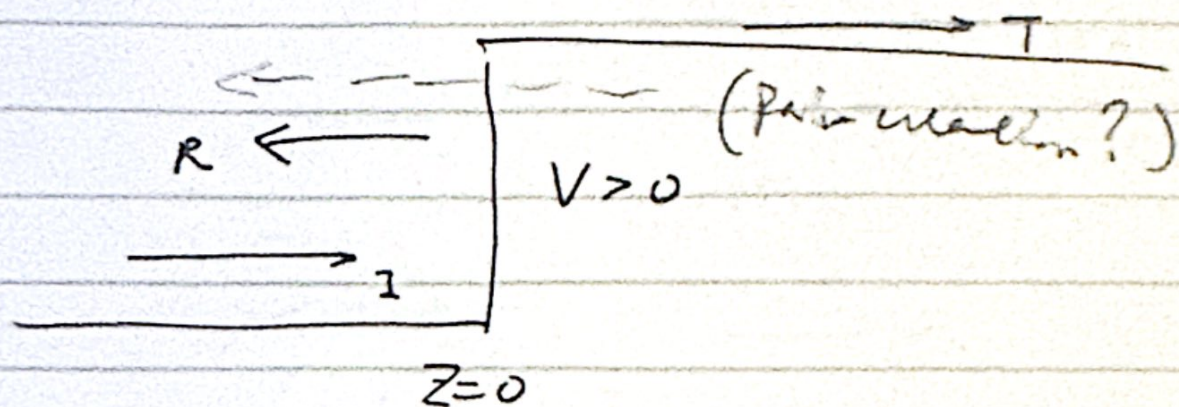
EM Field & Dirac

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + ieA_\mu$$

$$(-i\gamma^\mu D_\mu + m) \psi = 0$$

$$\underline{(-i\not{D} + m) \psi = 0}$$

same Gauge transformation as KG.



$$E^2 = p_z^2 + m^2 \quad z < 0$$

$$(E - eV)^2 = p_z'^2 + m^2 \quad z \geq 0$$

just match

$$e^{ip_z z} \begin{pmatrix} E+m \\ p_z \end{pmatrix} + r e^{-ip_z z} \begin{pmatrix} E+m \\ -p_z \end{pmatrix} \Big|_{z=0} = A e^{ip'_z z} \begin{pmatrix} E+m-eV \\ p'_z \end{pmatrix} \Big|_{z=0}$$

$$\frac{1-r}{1+r} \frac{p_z}{E+m} = \frac{p'_z}{E+m-eV}$$

$$\frac{1-r}{1+r} = K \left(= \frac{p'_z (E+m)}{p_z (E-eV+m)} \right)$$

$$r = \frac{1-K}{1+K}$$

$$A = \frac{2}{1+K}$$

3 Regimes

① $E > eV + m$

pass over

$$R = \left(\frac{1-K}{1+K} \right)^2 \quad T = \frac{4K}{(1+K)^2}$$

$$R + T = 1$$

② $E : eV - m < E < eV + m$

p'_z imag. $|r|^2 = 1$ R'' cannot pass.

$$\textcircled{3} \quad m < E < eV - m \quad P_z' \text{ real again}$$

$$R > 1, \quad T < 0 \quad (R+T=1)$$

Pair creation at boundary.

particles move left.

antiparticles move right

Dirac Eqn & Magnetic Dipole Moment

$$(-i\not{D} + m)\psi = 0$$

acting by $(i\not{D} + m)$:

$$(\gamma^\mu \gamma^\nu D_\mu D_\nu + m^2)\psi = 0$$

$$\begin{aligned} \gamma^\mu \gamma^\nu &= \frac{1}{2} \{ \gamma^\mu, \gamma^\nu \} + \frac{1}{2} [\gamma^\mu, \gamma^\nu] \\ &= g^{\mu\nu} + \Sigma^{\mu\nu} \end{aligned}$$

$$\underbrace{(D^\mu D_\mu + m^2)}_{\text{K-G, spinless}} \psi + \underbrace{\Sigma^{\mu\nu} D_\mu D_\nu}_{\text{modification for spin-1/2}} \psi = 0$$

K-G, spinless

modification
for spin-1/2.

$\Sigma^{\mu\nu}$ anti-symmetric.

$$D_\mu D_\nu = \underbrace{\frac{1}{2} \{D_\mu, D_\nu\}}_{\text{for which we need } \Sigma^{\mu\nu}} + \frac{1}{2} [D_\mu, D_\nu]$$

for which we need $\Sigma^{\mu\nu}$

$$\text{so } (D_\mu D^\mu + m^2 + \Sigma^{\mu\nu} \frac{1}{2} [D_\mu, D_\nu]) \psi = 0$$

$$[D_\mu, D_\nu] = ie(\partial_\mu A_\nu - \partial_\nu A_\mu) = ie F_{\mu\nu}$$

$$\text{take } \underline{A} = (-By, 0, 0) \quad \underline{B} = \text{and } \underline{A} = (0, 0, B)$$

$$A_\mu = (0, By, 0, 0)$$

$$F_{12} = -F_{21} = (\partial_1 A_2 - \partial_2 A_1) = -B$$

$$\Sigma^{12} = \frac{1}{2} (\gamma^1 \gamma^2 - \gamma^2 \gamma^1)$$

$$= -2i S_3 \longrightarrow \frac{1}{2} \begin{pmatrix} \sigma_3 & \\ & \sigma_3 \end{pmatrix}$$

$$0 = \left(D^\mu D_\mu + m^2 + \frac{ie}{2} (-B) (-2i S_3) \right) \psi$$

$\uparrow$
'12' & '21'

for (122).

$$0 = (D^\mu D_\mu + m^2 - 2eBS_3) \psi$$

Consider $E = m + \varepsilon$ ($\varepsilon/m \ll 1$)

$$\psi = e^{-it(m+\varepsilon)} \phi(\underline{x})$$

$$0 = \left(-(m+\varepsilon)^2 - (\nabla + ie\underline{A})^2 + m^2 - 2e\underline{B} \cdot \underline{S} \right) \phi$$

to 1st order in ε :

$$2\varepsilon m \phi = -(\nabla + ie\underline{A})^2 \phi - 2e\underline{B} \cdot \underline{S} \phi$$

$$\varepsilon \phi = -\frac{1}{2m} (\nabla + ie\underline{A})^2 \phi - \frac{e}{m} \underline{B} \cdot \underline{S} \phi$$

$$\underline{\mu}_S = \frac{e}{m} \underline{S} = 2\mu_B \underline{S}$$

$$\left(\mu_B = \frac{e}{2m} \right)$$

$$g_S = 2$$

More on S18

$$\langle x | p \rangle = \frac{e^{i p \cdot x / \hbar}}{(2\pi\hbar)^{3/2}}$$

Given $V|\psi_+\rangle = T|\phi\rangle$

Lippman-Schwinger: $|\psi_+\rangle = |\phi\rangle + (E - H_0 + i\varepsilon)^{-1} V|\psi_+\rangle$

$$T = V + V(E - H_0 + i\varepsilon)^{-1} T$$

$\sim$ 1st term

matrix element:

$$\langle \underline{p}_f | T | \underline{p}_i \rangle \approx \langle \underline{p}_f | \left(\int d^3x |\underline{x}\rangle \langle \underline{x}| \right) V | \underline{p}_i \rangle$$

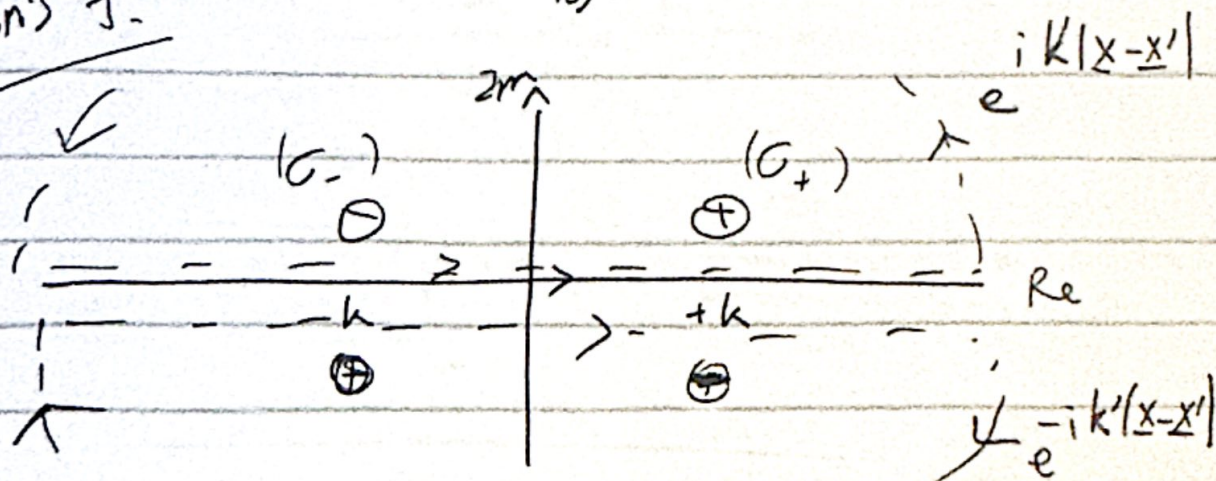
$$= \int d^3x \langle \underline{p}_f | \underline{x} \rangle V(\underline{x}) \langle \underline{x} | \underline{p}_i \rangle$$

$$= \frac{1}{(2\pi\hbar)^3} \int d^3x e^{i(\underline{p}_i - \underline{p}_f) \cdot \underline{x} / \hbar} V(\underline{x})$$

$$= \frac{1}{(2\pi\hbar)^3} \left(\frac{2\pi\hbar^2}{\mu} \right) f_k(\underline{k}_i, \underline{k}_f)$$

$$= -\frac{1}{4\pi^2\hbar\mu} f_k(\underline{k}_i, \underline{k}_f)$$

Carey's f.



Partial Wave Expansion

Energy unaffected. Outgoing does change

To ensure amplitudes (amplitudes unchanged, can just do a phase shift)

$$e^{ikz} = \sum_{l=0}^{\infty} e^{i\frac{\pi}{2}l} \sqrt{4\pi(2l+1)} Y_{l0}(\theta, \phi) \frac{1}{2kr} \left(e^{ikr - i(l+1)\frac{\pi}{2}} + e^{-ikr + i(l+1)\frac{\pi}{2}} \right)$$

$$\psi = \sum_{l=0}^{\infty} e^{i\frac{\pi}{2}l} \sqrt{4\pi(2l+1)} Y_{l0}(\theta, \phi) \frac{1}{2kr} \left(e^{ikr - i(l+1)\frac{\pi}{2}} \underbrace{e^{i2\delta_l}}_{\text{added}} - e^{-ikr + i(l+1)\frac{\pi}{2}} \right)$$

$$\text{so } \psi - e^{ikz} = \frac{e^{ikr}}{r} \sum_{l=0}^{\infty} \frac{1}{k} e^{i\delta_l} \sin(\delta_l) \sqrt{4\pi(2l+1)} Y_{l0}(\theta, \phi)$$

$$\Rightarrow f_k(\theta, \phi) = \sum_{l=0}^{\infty} \frac{1}{k} e^{i\delta_l} \sin(\delta_l) \sqrt{4\pi(2l+1)} Y_{l0}(\theta, \phi)$$

$$\text{By } Y_{l,0}(\theta, \phi) = \sqrt{\frac{(2l+1)}{4\pi}} P_l(\cos\theta)$$

$$f_k(\theta, \phi) = \sum_{l=0}^{\infty} \frac{(2l+1)}{k} e^{i\delta_l} \sin(\delta_l) P_l(\cos\theta)$$

$$\sigma_{\text{tot}} = \int \frac{d\sigma}{d\Omega} d\Omega = \int |f_k(\theta, \phi)|^2 d\Omega$$

$$= \sum_{l=0}^{\infty} \frac{4\pi}{k^2} \sin^2(\delta_l) (2l+1)$$

(by orthogonality)

$$\text{Also note } f_k(0, \phi) = \sum_{l=0}^{\infty} \frac{(2l+1)}{k} e^{i\delta_l} \sin(\delta_l)$$

$$\text{Im}\{f_k(0, \phi)\} = \left(\frac{4\pi}{k}\right)^{-1} \sigma_{\text{tot}}$$

$$\boxed{Q_{\text{el}} = \frac{4\pi}{k} \text{Im}\{f_k(0, \phi)\}}$$

Optical Theorem.

Can read off

$$e^{i2\delta_l} = \frac{1 - i\alpha_l}{1 + i\alpha_l}$$

(see notes for detail)

$$\rightarrow \delta_l = -\arctan(\alpha_l)$$

Spinors
 $s = \pm$

(normalized to $2E_p$)

$$u^s(p) = \sqrt{E_p + m} \begin{pmatrix} \xi^s \\ \frac{\mathbf{p} \cdot \boldsymbol{\sigma}}{E_p + m} \xi^s \end{pmatrix} \quad (\text{particle})$$

$$v^s(p) = \sqrt{E_p + m} \begin{pmatrix} -\frac{\mathbf{p} \cdot \boldsymbol{\sigma}}{E_p + m} \xi^s \\ \xi^s \end{pmatrix} \quad (\text{anti-particle})$$

ξ^s : two-component spinor
normalized to 1

$$u_{\underline{1}}^{st} u_{\underline{1}}^s = v_{\underline{1}}^{st} v_{\underline{1}}^s = 2E_p$$

$$\begin{aligned} u^\dagger \gamma^0 u &= 2m \\ v^\dagger \gamma^0 v &= -2m \end{aligned}$$

are Lorentz invariant scalars.